ACoP 2026 hackathon · simulated data · rehearsal, Mon 12 Oct 2026

DM1 covariate analysis

— who responds, and is it real?

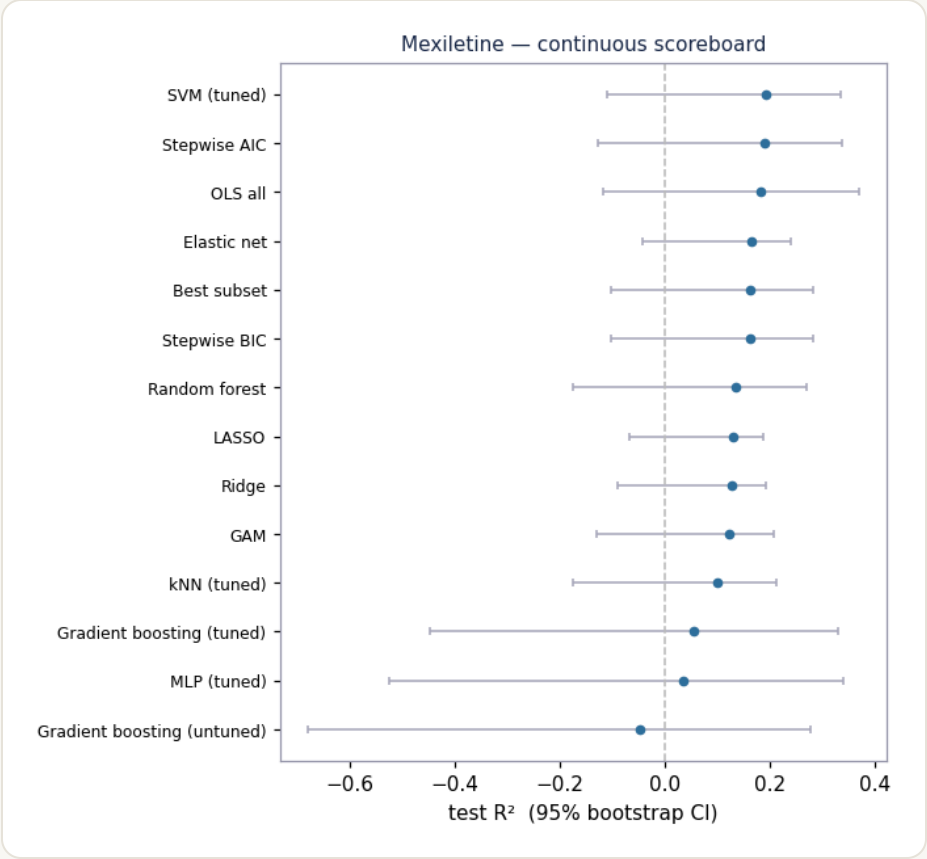
“scores move, covariates don’t”

Mexiletine **Oligonucleotide** **Tideglusib**

Held-out scoring traces to `outputs/report/slide_tables.md` and the files it cites. Marginal ρ from `results/tables/outcome_associations.csv`; the Mexiletine responder-prevalence enrichment is an in-sample descriptive statistic from the raw dataset (flagged on slides 6–7). The locked **fold 1** (n = 101, sha256 `8b3fb2cb…`, seed 2026) is **not** opened.

The hard part isn't *which* covariate predicts response — it's telling real signal from a lucky split

- **500 simulated DM1 patients, 3 therapies** (Mexiletine, Oligonucleotide, Tideglusib), **18 baseline covariates** each.
- One question per arm: *which baseline covariates predict who responds?* — asked on **two endpoints**, a continuous improvement score and a binary responder label.
- The trap: on data this size, a single train/test split can crown a different "answer" every time. The real question is which signals are **real** versus an artifact of who landed in the test set.
- Our claim: ask it the right way and the answer is **reproducible**; trust one leaderboard and it is **misleading**.

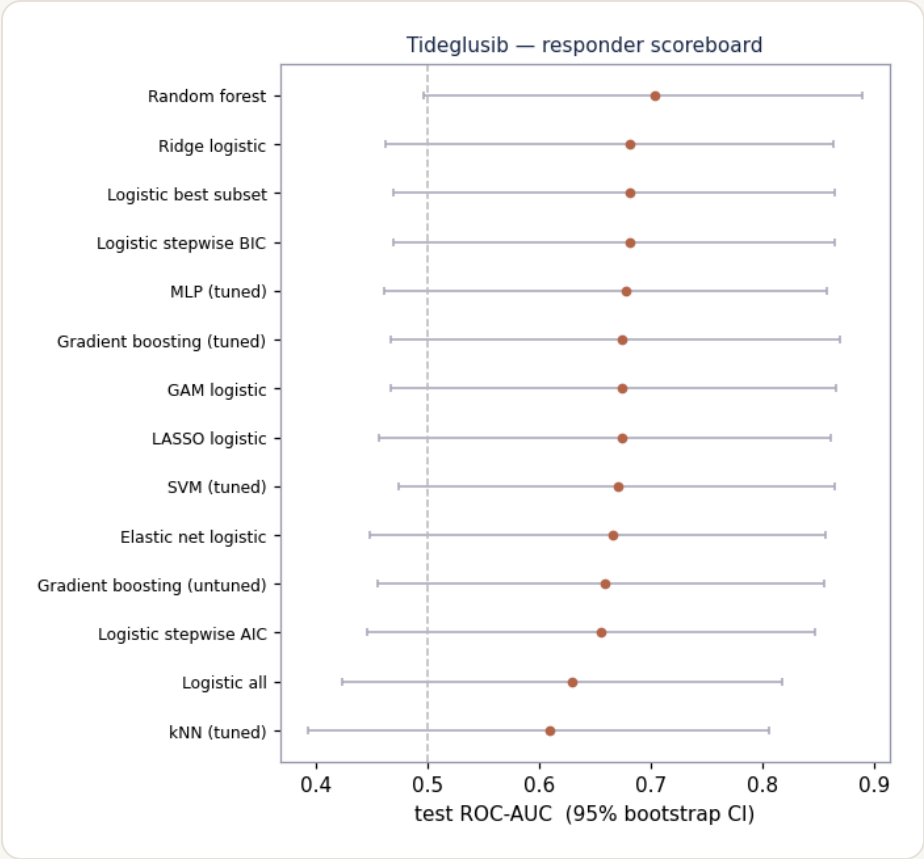


Mexiletine, continuous endpoint: all 14 methods' 95% bootstrap CIs cross zero — none is distinguishable from no-skill. **Which of these, if any, is real?**

Same held-out patients for all 14 models — so a score gap is the *method*, not the *sample*

14 models × 2 endpoints × 3 arms · one locked pipeline

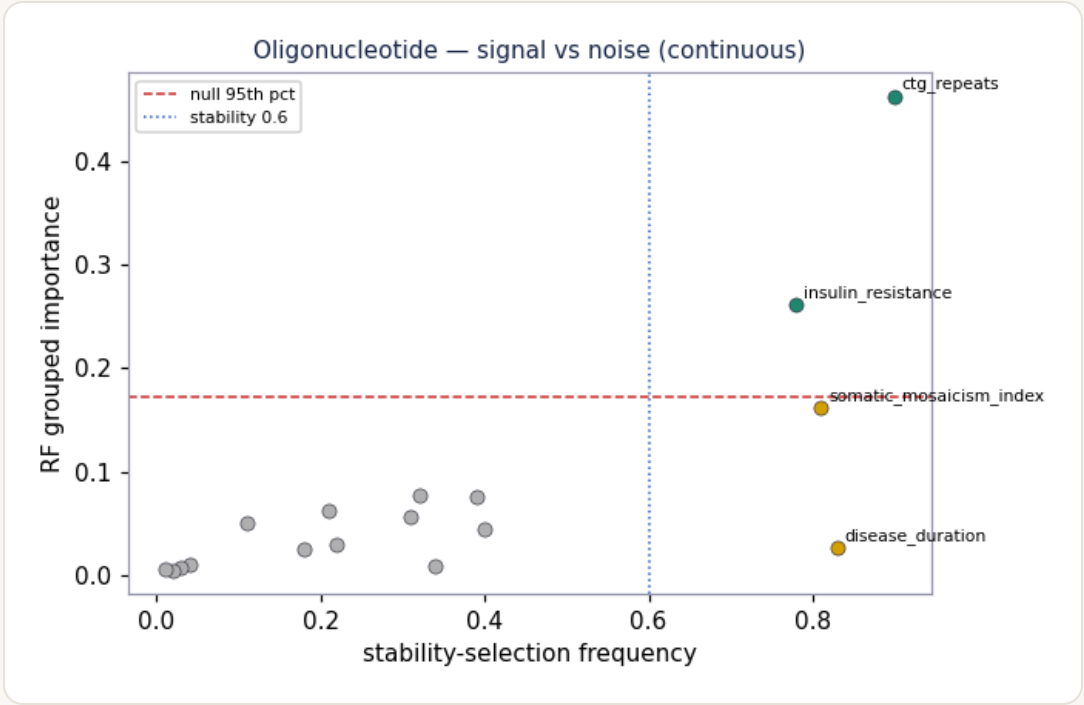
- **14 methods per task** — 8 inference (OLS/logistic, stepwise AIC, stepwise BIC, best subset, LASSO, elastic net, ridge, GAM) + 6 prediction (random forest, gradient boosting untuned/tuned, SVM, kNN, MLP).
- Gradient boosting = scikit-learn **HistGradientBoosting**, filling XGBoost's role (no OpenMP runtime here) — same job, fully reproducible (`docs/LOG.md`).
- All 14 scored on the **same held-out rows**, so a score gap is the *method*, not the *sample*.
- The real test set — **fold 1, 101 patients, sha256 `8b3fb2cb…`**, hash recorded Oct 11 — opens **exactly once** at the final run (G3). Everything today is rehearsal on fold 5; the discipline was fixed before we looked (`docs/hypothesis_sheet.md`).



The second endpoint (binary), 14 methods — all CIs overlap. **The ranking is a tie.**

The honest headline: the train/test split moves the *scores*, not the *covariates*

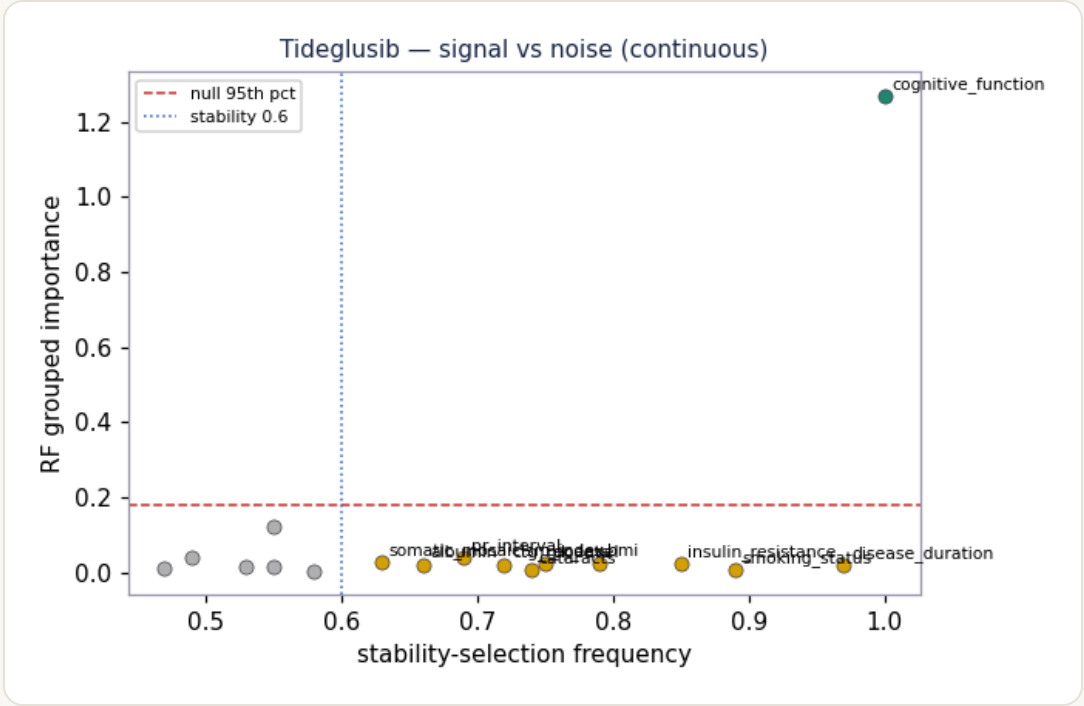
- Across **20 reseeded splits**, the median per-method test-R^2 SD is **0.131** (`compare/verdict.md`) — any ranking **inside that band is a tie**, not a winner.
- Same arm, different split, the leading R^2 swings: **Tideglusib +0.26 / +0.30 / +0.35; Mexiletine +0.19 / –0.01 / +0.14** (fold5 / single80 / single75).
- Yet the covariates barely move: across all 20 seeds, **Tideglusib cognition 100%, Oligonucleotide CTG repeats 90%, Mexiletine myotonia 85%**.
- So we never crown a model. We identify the **covariate signal that survives resampling** — that is the reproducible object.



CTG repeats (top-right): stable in ~90% of **100 half-subsamples** *(x-axis) and above the shuffled-RF null (dashed).* insulin_resistance *also clears here but fails cross-seed (17%) — see slide 4.*

A shuffled-outcome null plus stability selection separates *real signal* from a field of noise

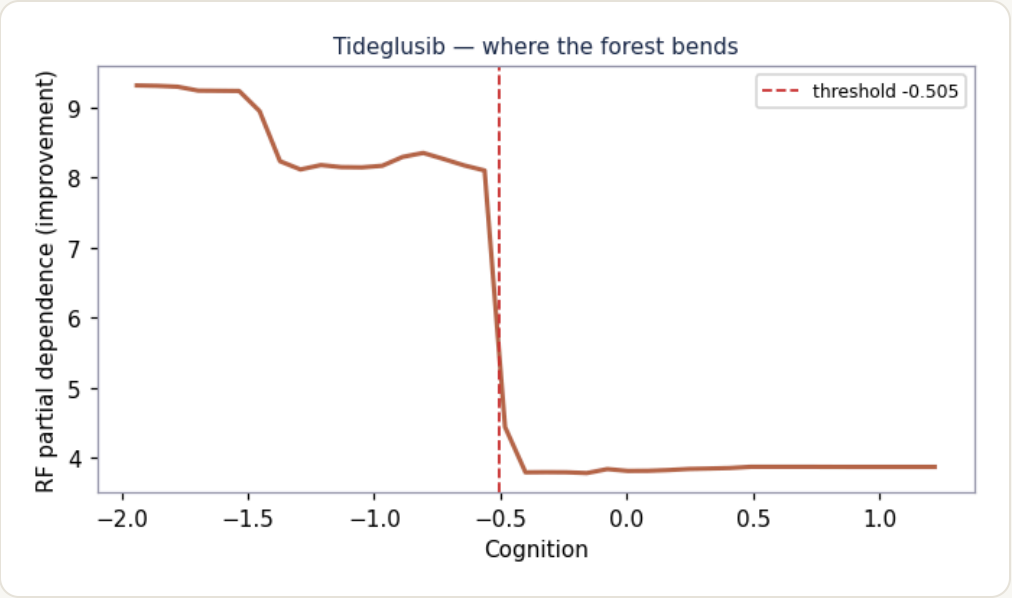
- **Pre-registered bar, fixed before scoring:** a covariate is **"real signal"** only if stable in **≥60% of 100 half-subsamples AND** beats the 95th percentile of a shuffled-outcome RF null (`docs/hypothesis_sheet.md`).
- **Replicating real signals:** Tideglusib cognition (both endpoints) and Oligonucleotide CTG repeats (both endpoints).
- **Honesty on display:** Mexiletine's severity trio (myotonia, CTG, duration) is stable but **inside the RF null** on the relative target → **"suggestive," not real** (`slide_tables.md`).
- **A flag we kept, not hid:** Oligo `insulin_resistance` met the real-signal bar on fold 5 (stable 0.78, beats the null 0.262 > 0.172) — but we override it: it holds in only **17% of cross-seed splits** (and q = 0.075 under FDR). **Downgraded, not claimed.**



Only **cognition (green)** *clears the dashed null; all 17 other covariates sit inside it — the overlapping labels along the bottom are all "suggestive" or noise.*

Tideglusib is the showcase: ML found a cognition threshold, a one-term rule captures it, and biology explains it

- **ML found it:** random forest's **20-seed mean R^2 = 0.491** vs **~0.29 (OLS) – 0.36 (LASSO)** for the linear models (`compare/repeat20_r2.csv`) — a nonlinearity linear fits miss.
- **It's a step, not a slope:** partial dependence holds a high plateau (~8) for low baseline cognition, then **cliffs ~4 points** to a ~3.8 floor and flattens — a spline GAM (R^2 0.119) underfits it.
- **One term captures it:** the data-chosen cut `1[cognitive_function ≤ −0.505]` lifts single-term test R^2 from **+0.233 (linear) to +0.377 (indicator)** (`rule.txt`) — one held-out fold (~34 pts), so it *corroborates* the 20-seed + cross-seed evidence; re-estimated on fold 1.
- **Biology explains it:** a cognition-weighted composite + GSK-3β target → a head-room/floor effect. In-sample (48/102) the rule enriches mean improvement **3.46 → 8.28 (2.4×)**, keeping ~47% — but with **no placebo arm this is on-treatment response**, i.e. **prognostic** enrichment to confirm against a control, not yet separable from regression-to-the-mean / floor. Marginal ρ **−0.647** (q = 4e-16; `outcome_associations.csv`); the pre-registered step near z ≈ −0.5 landed on **−0.505**.



RF partial dependence: a ~4-point cliff (≈8 → ≈3.8) at the data-chosen cut **cognition = −0.505** *(dashed) — a step, not a slope.*

Both other arms, told honestly: the label/axis is *real*; the per-patient magnitude isn't

Mexiletine — label yes, magnitude no

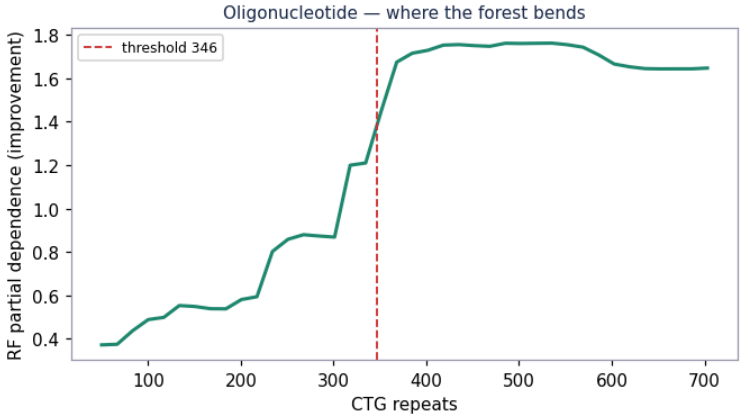
- The responder **label** is cleanly separable (stepwise logistic **ROC-AUC 0.773, CI 0.587–0.928**), even though improvement **magnitude** is not (best $R^2 \approx 0.19$, CI through 0). The relative target divides severity back out.
- **Enrichment (prognostic):** CTG > 448 ~doubles responder prevalence, **0.289 → 0.543** — an *in-sample descriptive stat* over folds 2–5 (N=128; enriched n=35, wide CI), from the raw dataset (`interpretation_Mexiletine.md` §6), *not* held-out scoring.

Oligonucleotide — axis yes, prediction no

- CTG repeats is a **real, stable axis** (90% on the continuous endpoint — half-subsample + cross-seed; 76% binary) but per-patient QMT change is **unpredictable out-of-sample** (R^2 0.057, CI [–0.29, +0.25]). Use CTG to **stratify, not predict individuals**.
- **Flag for the locked run:** repeat-burden direction is **positive** here (ρ +0.42, q = 6e-6; `outcome_associations.csv`), **opposite** the mechanistic ceiling prior — pre-specify the sign, confirm on fold 1.

AUC 0.773

Mexiletine responder label — top stepwise logistic, 95% CI 0.587–0.928; clears 0.5 (no-skill). The *label* is separable even though improvement *magnitude* is not.

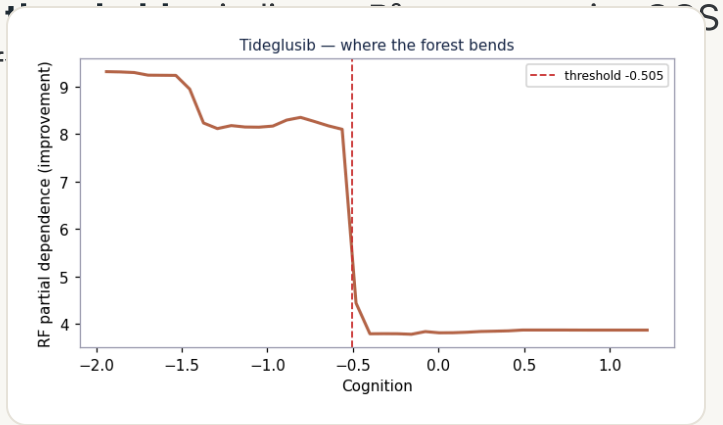


Oligonucleotide: *CTG partial dependence rises monotonically (0.38 → 1.76)* — **no exploitable step**.

The deliverable is three pre-registered enrichment rules — and the discipline that makes them trustworthy before the locked run

ARM	RULE (DATA-CHOSEN)	EFFECT (IN-SAMPLE, ON-TREATMENT)	STATUS
Tideglusib	enroll `cognition z ≤ −0.5`	8.28 vs 3.46, keeps ~47% (48/102)	**Adopted (prognostic)** — step beats the linear term out-of-sample (+0.377 > +0.233); backed by 20-seed + 100% cross-seed. On-treatment, **no placebo** → confirm against a control on fold 1.
Mexiletine	`CTG > 448`	responder prevalence 0.289 → 0.543 (folds 2–5, N=128; n=35 enriched)	**Prognostic on the label** — the magnitude cut does *not* beat the linear term (+0.138 < +0.163), so we don't claim it there. Pre-specify; confirm on fold 1.
Oligonucleotide	CTG as **stratifier only**	orders patients 2.03 vs 0.38 (41/102)	**No magnitude** (−0.071). Stratif

- Nothing here rests on a single lucky split: held-out scores carry **bootstrap CIs** and a **20-seed SD (0.131)**; signals carry a **cross-seed frequency**. (The enrichment effect sizes above are in-sample/training point estimates — wide CI, re-estimated on fold 1.)
- **Next (G3):** open the locked **fold 1** once, confirm hash `8b3fb2cb…`, refresh every number. The rules and the honesty are **already pre-registered**.



Callback to slide 5 — the one rule we would pre-register today: **enroll cognition z ≤ −0.5**.